

## NEUSIEDL AM SEE, Austria

WMO: 111940

Lat: 47.951N

Lon: 16.842E

Elev: 148

StdP: 99.56

Time Zone: 1.00 (EUC)

Period: 94-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.7       | -7.4 | -13.6                         | 1.2 | -8.7 | -10.8 | 1.5 | -6.0 | 9.5                   | 4.5  | 8.4 | 2.7  | 3.1                   | 330  | 0.469 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.2                   | 32.5            | 22.6 | 30.6 | 21.6 | 28.9 | 20.7 | 23.2                | 31.2 | 22.2 | 29.4 | 21.2 | 27.7 | 2.9                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.5                            | 15.5 | 27.5 | 19.6 | 14.6 | 26.2 | 18.8 | 13.8 | 25.1 | 69.6          | 31.6 | 65.6 | 29.5 | 62.1 | 27.8 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 8.1               | 7.0  | 6.2 | -11.9                      | 35.1 | 3.4                | 2.4 | -14.4                                              | 36.8 | -16.4      | 38.2 | -18.3      | 39.5 | -20.8      | 41.2 |  |  |
|                   |      |     | -12.4                      | 24.6 | 3.2                | 1.1 | -14.7                                              | 25.4 | -16.6      | 26.0 | -18.4      | 26.6 | -20.8      | 27.4 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.3        | 0.1   | 2.1   | 6.4   | 11.9  | 16.4  | 19.7  | 21.9  | 21.5  | 16.6  | 11.3  | 6.2   | 1.2   | (6)  |
|                                                                        | DBStd       | 8.62        | 4.43  | 4.56  | 4.24  | 4.25  | 3.58  | 3.62  | 3.53  | 3.31  | 3.50  | 4.03  | 4.10  | 4.17  | (7)  |
|                                                                        | HDD10.0     | 1117        | 306   | 222   | 126   | 29    | 1     | 0     | 0     | 0     | 1     | 34    | 124   | 274   | (8)  |
|                                                                        | HDD18.3     | 2897        | 564   | 454   | 369   | 193   | 82    | 27    | 8     | 9     | 76    | 220   | 363   | 532   | (9)  |
|                                                                        | CDD10.0     | 1603        | 1     | 1     | 14    | 88    | 198   | 291   | 370   | 357   | 197   | 73    | 12    | 1     | (10) |
|                                                                        | CDD18.3     | 342         | 0     | 0     | 0     | 2     | 22    | 68    | 120   | 107   | 22    | 1     | 0     | 0     | (11) |
|                                                                        | CDH23.3     | 2854        | 0     | 0     | 0     | 20    | 183   | 564   | 1072  | 879   | 131   | 6     | 0     | 0     | (12) |
|                                                                        | CDH26.7     | 958         | 0     | 0     | 0     | 2     | 32    | 170   | 416   | 317   | 21    | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 2.9         | 3.2   | 3.2   | 3.3   | 3.2   | 2.9   | 2.6   | 2.7   | 2.4   | 2.6   | 2.5   | 2.9   | 2.9   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 602         | 30    | 28    | 40    | 39    | 64    | 70    | 73    | 65    | 63    | 47    | 46    | 36    | (15) |
|                                                                        | PrecMax     | 786         | 68    | 72    | 85    | 84    | 146   | 152   | 199   | 160   | 166   | 96    | 98    | 71    | (16) |
|                                                                        | PrecMin     | 393         | 6     | 0     | 6     | 0     | 15    | 24    | 6     | 7     | 18    | 2     | 1     | 7     | (17) |
|                                                                        | PrecStd     | 89          | 17    | 20    | 24    | 22    | 32    | 33    | 43    | 37    | 42    | 26    | 21    | 17    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 13.4  | 15.3  | 20.3  | 25.8  | 29.6  | 32.8  | 35.0  | 35.0  | 29.1  | 23.9  | 18.4  | 12.9  | (19) |
|                                                                        |             | MCWB        | 8.6   | 9.5   | 12.6  | 16.3  | 20.1  | 22.6  | 24.2  | 22.9  | 20.6  | 17.4  | 13.5  | 9.2   | (20) |
|                                                                        | 2%          | DB          | 10.4  | 12.2  | 17.6  | 23.2  | 27.3  | 30.6  | 32.7  | 32.4  | 26.7  | 20.9  | 15.8  | 9.7   | (21) |
|                                                                        |             | MCWB        | 7.1   | 8.1   | 11.3  | 15.2  | 19.0  | 21.8  | 22.8  | 22.4  | 19.3  | 16.0  | 12.3  | 7.0   | (22) |
|                                                                        | 5%          | DB          | 8.3   | 10.1  | 15.0  | 21.2  | 25.4  | 28.7  | 30.8  | 30.1  | 24.7  | 19.1  | 13.7  | 7.6   | (23) |
|                                                                        |             | MCWB        | 5.5   | 7.0   | 9.9   | 14.2  | 17.8  | 20.7  | 21.8  | 21.4  | 18.1  | 15.1  | 10.9  | 5.5   | (24) |
|                                                                        | 10%         | DB          | 6.3   | 8.2   | 12.9  | 19.1  | 23.4  | 26.7  | 29.0  | 28.1  | 22.6  | 17.1  | 11.8  | 6.2   | (25) |
|                                                                        |             | MCWB        | 4.2   | 5.7   | 8.8   | 12.9  | 16.6  | 19.5  | 20.9  | 20.4  | 17.1  | 13.7  | 9.6   | 4.6   | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 9.0   | 10.0  | 13.2  | 17.3  | 21.1  | 23.9  | 24.8  | 24.4  | 21.1  | 17.9  | 14.1  | 9.5   | (27) |
|                                                                        |             | MCDB        | 12.1  | 14.5  | 19.1  | 24.9  | 28.4  | 31.4  | 33.6  | 32.8  | 27.5  | 22.8  | 17.6  | 12.3  | (28) |
|                                                                        | 2%          | WB          | 7.6   | 8.4   | 11.7  | 15.7  | 19.5  | 22.5  | 23.3  | 23.0  | 19.9  | 16.7  | 12.5  | 7.2   | (29) |
|                                                                        |             | MCDB        | 10.4  | 11.7  | 16.4  | 22.2  | 26.2  | 29.2  | 31.4  | 31.3  | 25.2  | 20.2  | 15.4  | 9.2   | (30) |
|                                                                        | 5%          | WB          | 5.7   | 7.2   | 10.3  | 14.6  | 18.3  | 21.2  | 22.3  | 21.7  | 18.9  | 15.4  | 11.2  | 5.9   | (31) |
|                                                                        |             | MCDB        | 7.7   | 9.9   | 14.3  | 20.2  | 24.4  | 27.4  | 29.7  | 28.9  | 23.6  | 18.2  | 13.6  | 7.3   | (32) |
|                                                                        | 10%         | WB          | 4.3   | 5.9   | 9.1   | 13.5  | 17.2  | 20.1  | 21.3  | 20.8  | 17.7  | 14.3  | 9.7   | 4.7   | (33) |
|                                                                        |             | MCDB        | 6.0   | 8.0   | 12.5  | 18.3  | 22.3  | 25.7  | 27.9  | 27.1  | 21.7  | 16.6  | 11.5  | 6.0   | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.6   | 6.2   | 7.9   | 9.7   | 9.9   | 10.0  | 10.2  | 10.0  | 8.5   | 7.4   | 5.1   | 4.1   | (35) |
|                                                                        |             | MCDBR       | 7.1   | 9.3   | 11.7  | 12.6  | 12.9  | 12.7  | 13.0  | 13.4  | 11.6  | 9.7   | 7.9   | 6.6   | (36) |
|                                                                        | 5% WB       | MCWBR       | 4.7   | 6.0   | 6.8   | 6.5   | 6.0   | 5.4   | 5.2   | 5.5   | 5.4   | 5.2   | 5.0   | 4.8   | (37) |
|                                                                        |             | MCWBR       | 6.6   | 8.7   | 11.0  | 11.9  | 12.3  | 12.0  | 12.4  | 12.7  | 10.8  | 8.9   | 7.6   | 6.1   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.321 | 0.345 | 0.382 | 0.426 | 0.423 | 0.426 | 0.434 | 0.434 | 0.408 | 0.378 | 0.352 | 0.320 | (40) |
|                                                                        |             | taud        | 2.379 | 2.349 | 2.267 | 2.168 | 2.224 | 2.251 | 2.231 | 2.246 | 2.288 | 2.344 | 2.372 | 2.402 | (41) |
|                                                                        | Ebn at Noon |             | 744   | 806   | 827   | 824   | 842   | 840   | 826   | 808   | 791   | 753   | 695   | 697   | (42) |
|                                                                        |             | Edh at Noon | 70    | 89    | 113   | 138   | 135   | 133   | 133   | 126   | 110   | 89    | 71    | 61    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.01  | 1.82  | 3.02  | 4.51  | 5.33  | 5.81  | 5.63  | 5.01  | 3.58  | 2.20  | 1.14  | 0.78  | (44) |
|                                                                        | RadStd      |             | 0.11  | 0.25  | 0.39  | 0.44  | 0.56  | 0.42  | 0.49  | 0.48  | 0.43  | 0.24  | 0.11  | 0.09  | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (83 neighbors) | +0.57 | N/A     | N/A    | +0.86   | +0.47 | +0.42 | -120        | -165    | +91     | +47     |

Nomenclature: See separate page